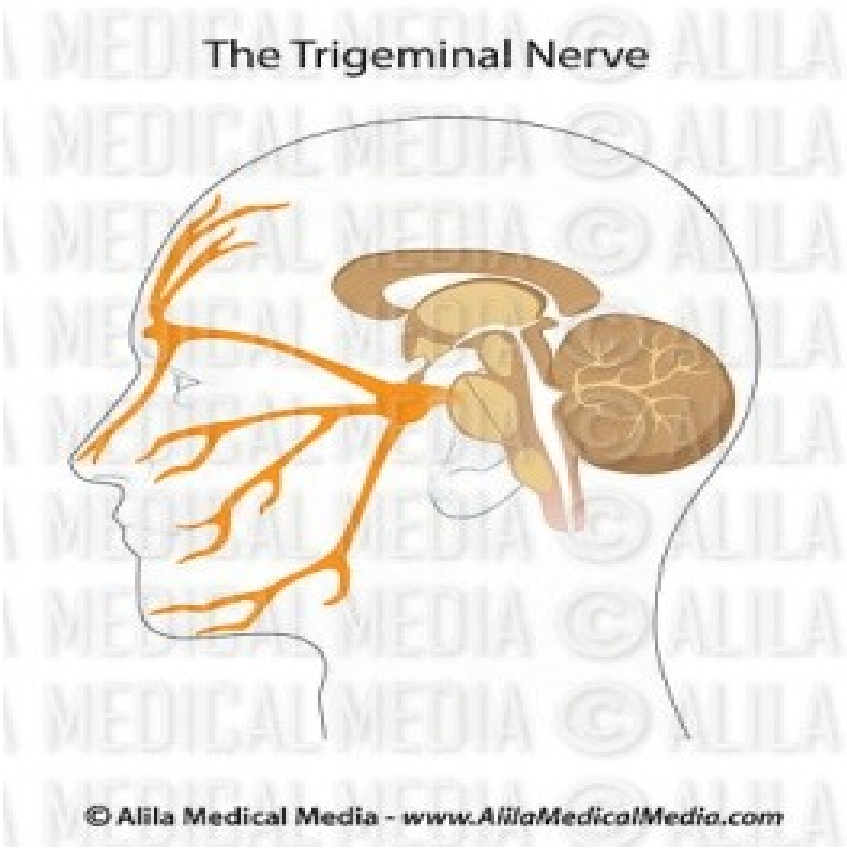
The Trigeminal Nerve (Cranial Nerve V)



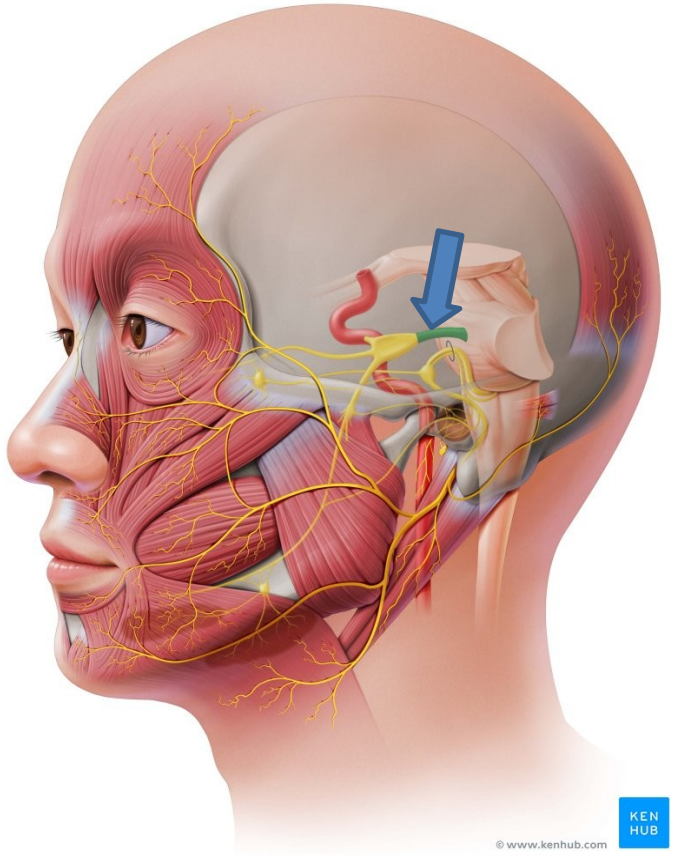
Learning Objectives:

At the end of the lecture, the students should be able to:

- **Define site of origin & type of trigeminal nerve.**
- **Enlist nuclei & main branches/divisions and functional components of nerve.**
- **Describe areas of distribution & main course of each of its branches.**
- **Define trigeminal neuralgia.**

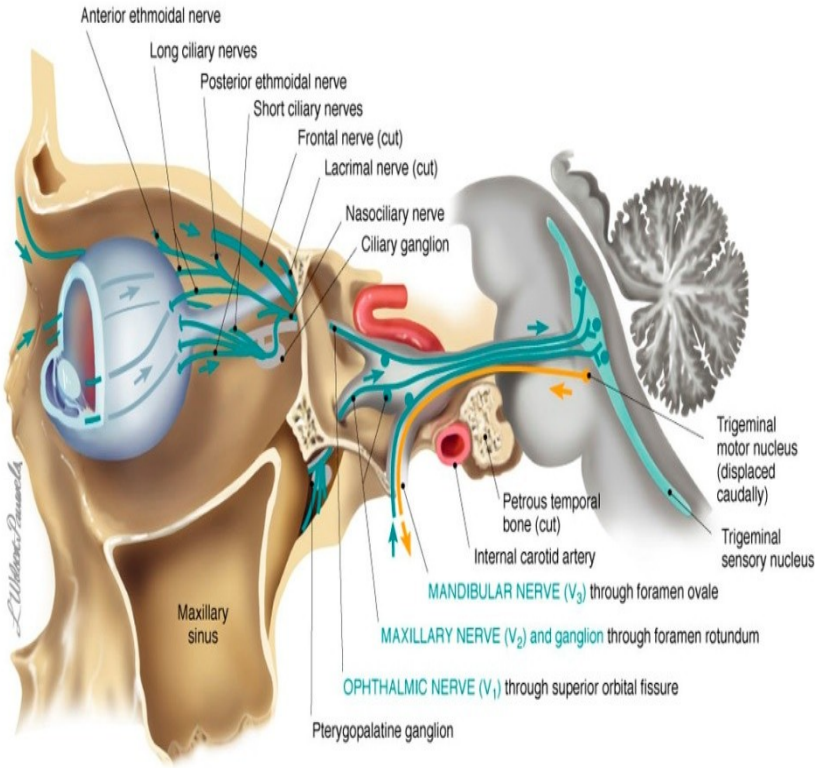
Introduction:

- **The trigeminal nerve is named so because it has 3 main divisions i.e. Ophthalmic, Maxillary and Mandibular nerves.**
- **It is fifth cranial nerve and largest of all cranial nerves.**
- **It is a mixed nerve i.e. having both sensory and motor components.**
- **It is sensory to greater part of head, teeth and oral and nasal cavities & motor to some muscles such as muscles of mastication.**



Origin & course:

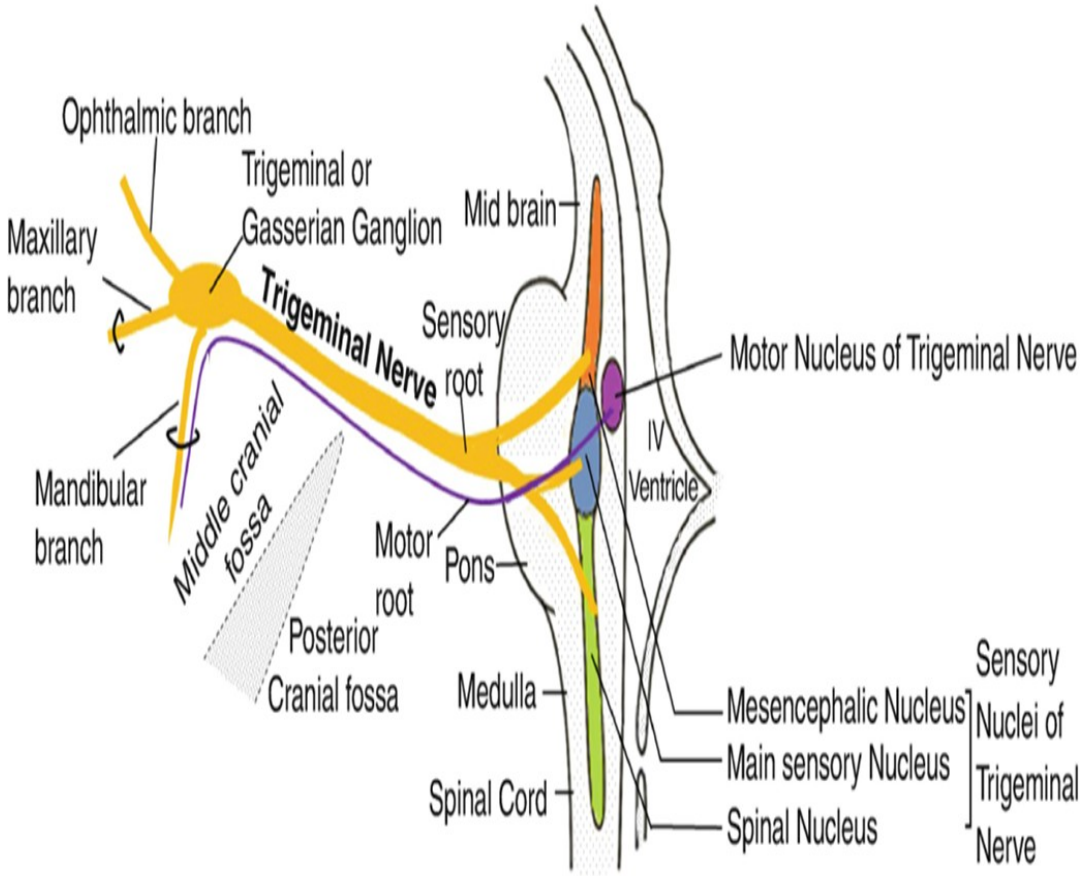
- It arises from the ventral aspect of pons as a small motor and a large sensory root.
- It runs forwards out of posterior cranial fossa and rests on the apex of petrous temporal bone in middle cranial fossa.
- The larger sensory root here expands to form a crescent-shaped trigeminal ganglion which lies in a pouch of dura mater called meckel cave.



Nuclei:

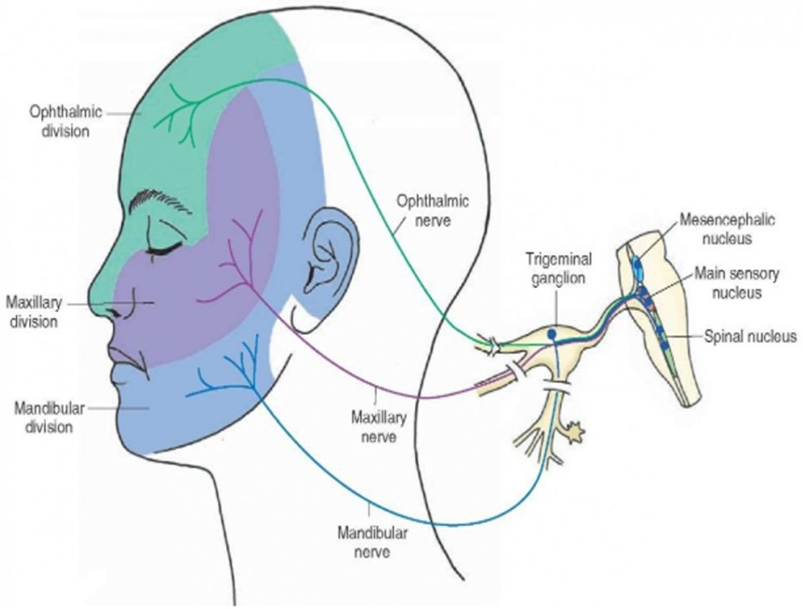
The trigeminal nerve has four nuclei:

1) **The main sensory nucleus**
2) **The spinal nucleus**
3) **The mesencephalic nucleus &**
4) **The motor nucleus**



Branches:

The ophthalmic, maxillary & mandibular branches of trigeminal nerve arise from anterior border of trigeminal ganglion.

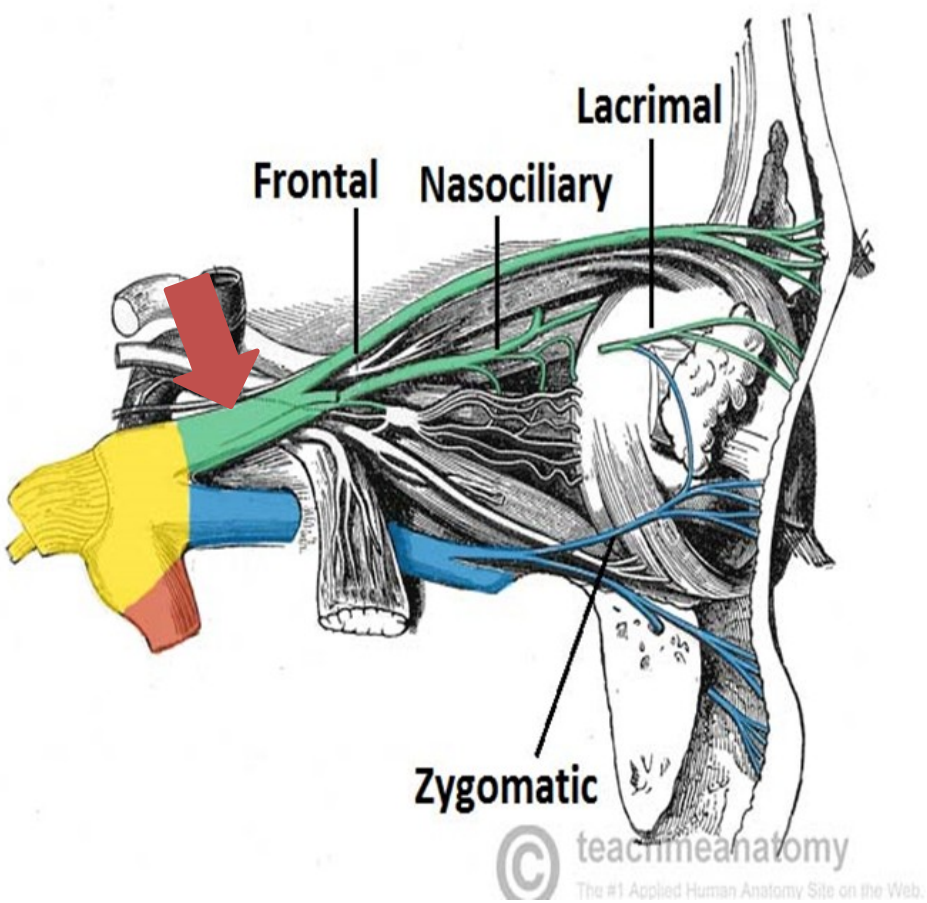


Course of Trigeminal Nerve

https://www.medicosite.com/

Ophthalmic nerve (V1):

- **The ophthalmic division is sensory**
- **It leaves the skull through superior orbital fissure to reach the orbit.**
- **It has three terminal branches viz, frontal, lacrimal & nasociliary nerves.**



Distribution of ophthalmic nerve:

REFERENCE HANDOUT: TRIGEMINAL NERVE BRANCHES (NOT INCLUDING HITCHHIKING PATHWAYS OF VII, IX) zill@musom.2013

V1 Ophthalmic - GSA (Sensory) only - through Superior Orbital Fissure to Orbit

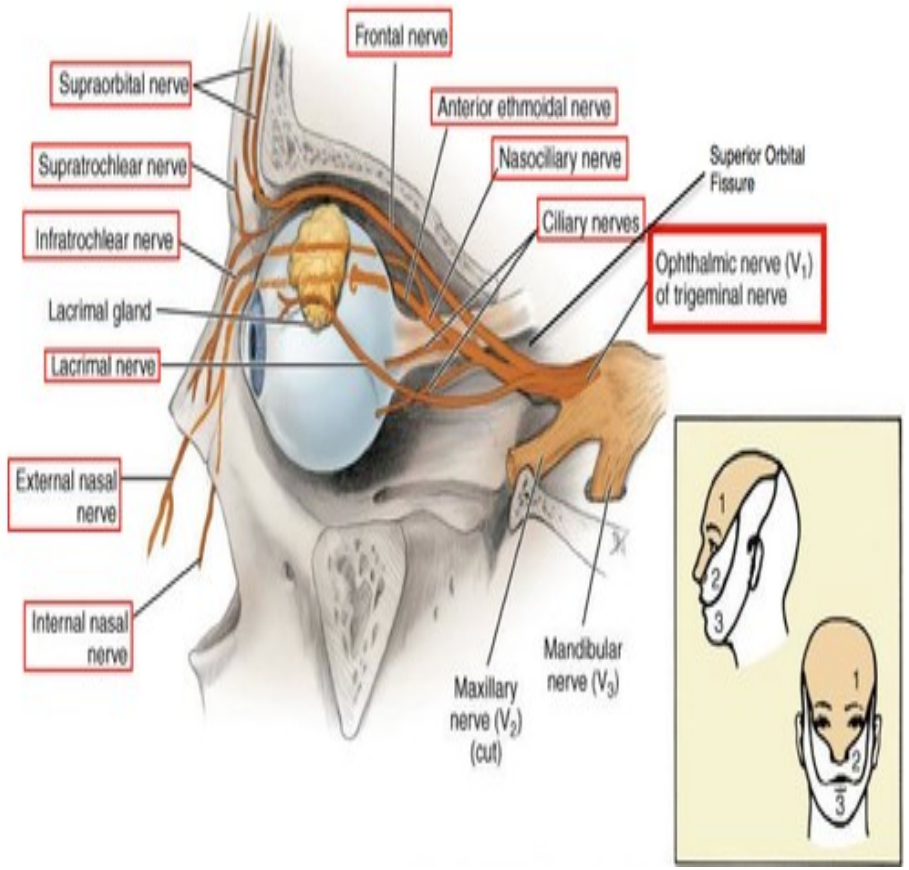
Nerve	Branches	Innervates
1. Frontal Nerve	a. Supraorbital Nerve	Scalp forehead, upper eyelid
	b. Supratrochlear Nerve	Scalp forehead, upper eyelid
2. Lacrimal Nerve		Upper eyelid
3. Nasociliary Nerve	a. Long Ciliary Nerve	Cornea of eye
	b. Ant. and Post. Ethmoidal Nerves	Nasal cavity, ethmoid sinus, tip of nose
	c. Infratrochlear Nerve	Upper eyelid, nose

V2 Maxillary - GSA (Sensory) only - through Foramen Rotundum to Pterygopalatine Fossa

Nerve	Branches	Innervates
1. Meningeal branches		Dura of mid. Cranial fossa
2. Ganglionic branches	a. Greater Palatine Nerve	Hard Palate
	b. Lesser Palatine Nerve	Soft Palate
	c. Nasopalatine Nerve	Nasal Cavity, Hard Palate
	d. Nasal branches	Nasal Cavity
3. Post. Sup. Alveolar Nerve		Maxillary teeth
4. Infraorbital nerve		Lower eyelid, nose, upper lip
	a. Ant. Sup. Alveolar Nerve	Maxillary teeth
	b. Mid. Sup. Alveolar Nerve	Maxillary teeth
5. Zygomatic nerve	a. Zygomaticofacial Nerve	Skin of cheek
	b. Zygomaticotemporal Nerve	Skin of temporal region

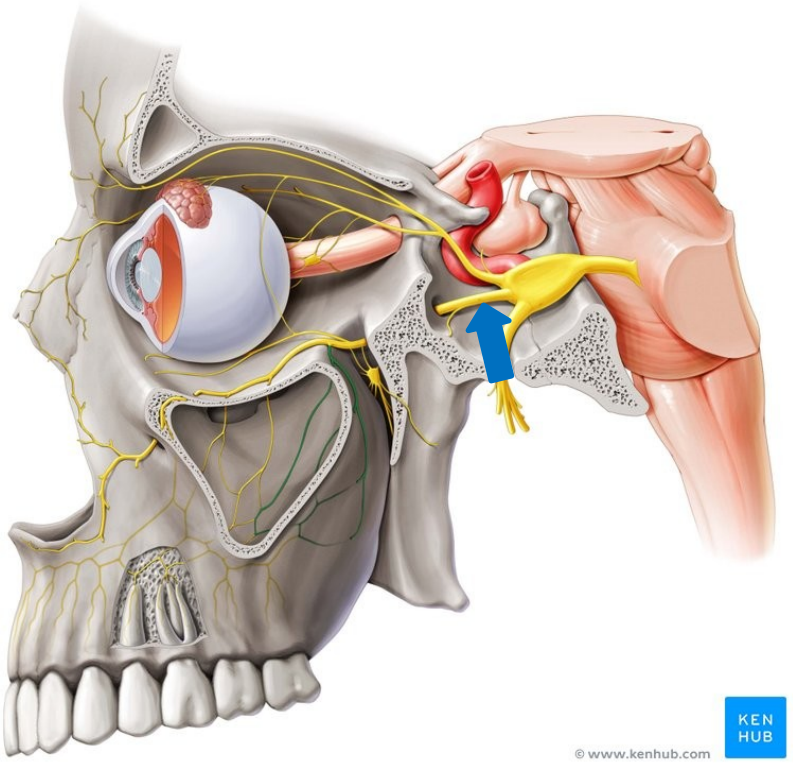
V3 Mandibular - GSA (Sensory) and SVE (Motor) - Foramen Ovale to Infratemporal Fossa

Nerve	Branches	Innervates
1. Nervous spinosus		Sensory to Dura of mid Cranial fossa
2. Motor branches		Motor to Med. Pterygoid, Tens. Tympani, Tensor Palati
3. Anterior division	a. Nerve to Lateral Pterygoid	Motor to Lateral Pterygoid
	b. Masseteric Nerve	Motor to Masseter
	c. Deep Temporal Nerve	Motor to Temporalis
	d. Buccal Nerve	Sensory to Cheek
4. Posterior Division	a. Auriculotemporal Nerve	Sensory to external auditory meatus, tympanic membrane, TMJ, lateral scalp
	b. Lingual Nerve	Sensory (touch) ant. 2/3 tongue
	c. Inferior Alveolar Nerve i. Nerve to Mylohyoid ii. Mental Nerve	Sensory to Mandibular teeth Motor to Mylohyoid, ant. Digastric Sensory to Chin, Lower lip



Maxillary nerve (V2):

- **It is also sensory.**
- **It leaves the skull through foramen rotundum to reach the pterygopalatine fossa.**
- **It gives meningeal branches, branches through pterygopalatine ganglion and zygomatic & posterior superior alveolar branches and beyond pterygopalatine fossa it continues as infraorbital nerve.**



Distribution of maxillary nerve:

REFERENCE HANDOUT: TRIGEMINAL NERVE BRANCHES (NOT INCLUDING HITCHHIKING PATHWAYS OF VII, IX) zill@musom.2013

V1 Ophthalmic - GSA (Sensory) only - through Superior Orbital Fissure to Orbit

Nerve	Branches	Innervates
1. Frontal Nerve	a. Supraorbital Nerve	Scalp forehead, upper eyelid
	b. Supratrochlear Nerve	Scalp forehead, upper eyelid
2. Lacrimal Nerve		Upper eyelid
3. Nasociliary Nerve	a. Long Ciliary Nerve	Cornea of eye
	b. Ant. and Post. Ethmoidal Nerves	Nasal cavity, ethmoid sinus, tip of nose
	c. Infratrochlear Nerve	Upper eyelid, nose

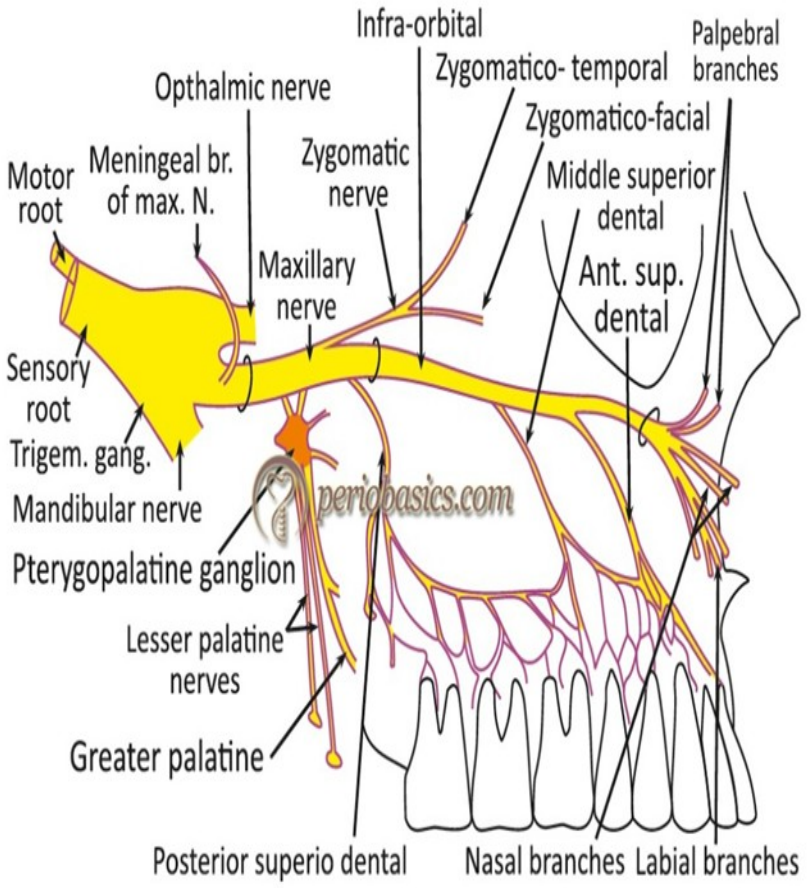
V2 Maxillary - GSA (Sensory) only - through Foramen Rotundum to Pterygopalatine Fossa

Nerve	Branches	Innervates
1. Meningeal branches		Dura of mid. Cranial fossa
2. Ganglionic branches	a. Greater Palatine Nerve	Hard Palate
	b. Lesser Palatine Nerve	Soft Palate
	c. Nasopalatine Nerve	Nasal Cavity, Hard Palate
	d. Nasal branches	Nasal Cavity
3. Post. Sup. Alveolar Nerve		Maxillary teeth
4. Infraorbital nerve		Lower eyelid, nose, upper lip
	a. Ant. Sup. Alveolar Nerve	Maxillary teeth
	b. Mid. Sup. Alveolar Nerve	Maxillary teeth
5. Zygomatic nerve	a. Zygomaticofacial Nerve	Skin of cheek
	b. Zygomaticotemporal Nerve	Skin of temporal region

V3 Mandibular - GSA (Sensory) and SVE (Motor) - Foramen Ovale to Infratemporal Fossa

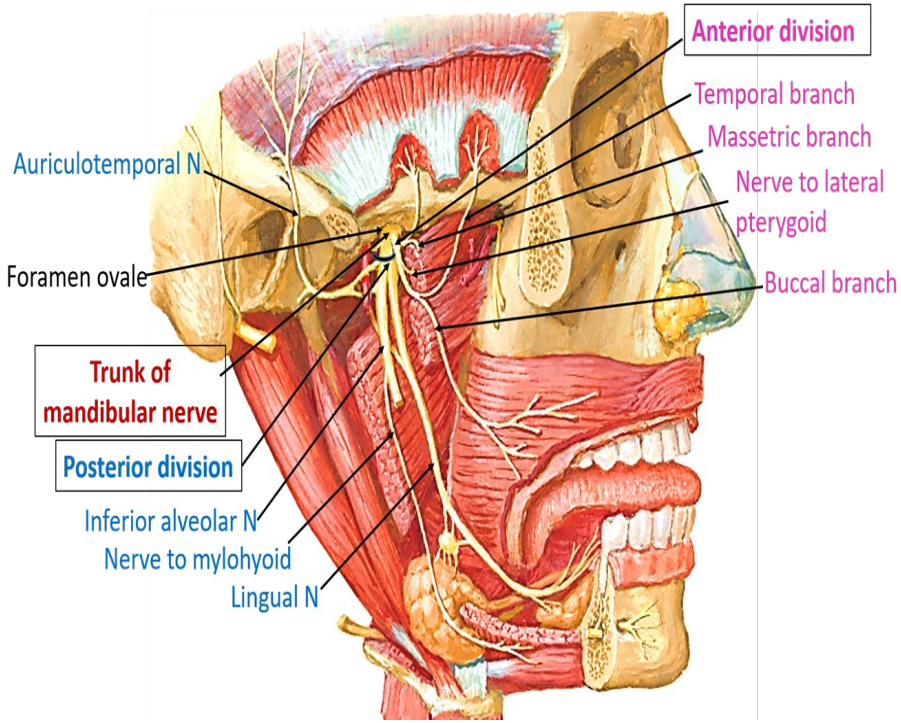
Nerve	Branches	Innervates
1. Nervous spinosus		Sensory to Dura of mid Cranial fossa
2. Motor branches		Motor to Med. Pterygoid, Tens. Tympani, Tensor Palati
3. Anterior division	a. Nerve to Lateral Pterygoid	Motor to Lateral Pterygoid
	b. Masseteric Nerve	Motor to Masseter
	c. Deep Temporal Nerve	Motor to Temporalis
	d. Buccal Nerve	Sensory to Cheek
4. Posterior Division	a. Auriculotemporal Nerve	Sensory to external auditory meatus, tympanic membrane, TMJ, lateral scalp
	b. Lingual Nerve	Sensory (touch) ant. 2/3 tongue
	c. Inferior Alveolar Nerve i. Nerve to Mylohyoid ii. Mental Nerve	Sensory to Mandibular teeth Motor to Mylohyoid, ant. Digastric Sensory to Chin, Lower lip

1



Mandibular nerve (V3):

- It is the largest of the three divisions of trigeminal nerve
- It is the nerve of first branchial arch and supplies all the structures derived from it.
- It is mixed i.e. containing both sensory & motor roots.
- The sensory root arises from lateral aspect of trigeminal ganglion & motor root lies deep to it.
- Both the roots leave the cranial cavity through foramen ovale. Just below foramen both roots combine to form single main trunk which lies in infratemporal fossa.
- The main trunk divides into a smaller anterior and a large posterior division.



Divisions of mandibular nerve:

- From the main trunk:

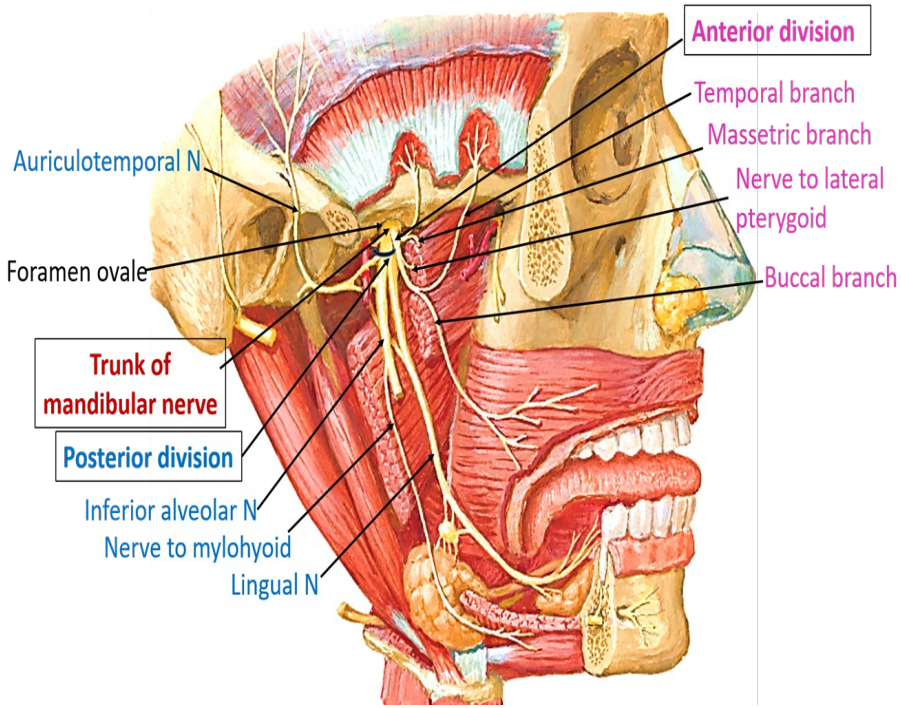
1) Meningeal branch &
2) Nerve to medial pterygoid.

- From anterior division:

1) Sensory branch, buccal nerve &
2) Motor branches viz, massetric & deep temporal nerves and nerve to lateral pterygoid.

- From posterior division:

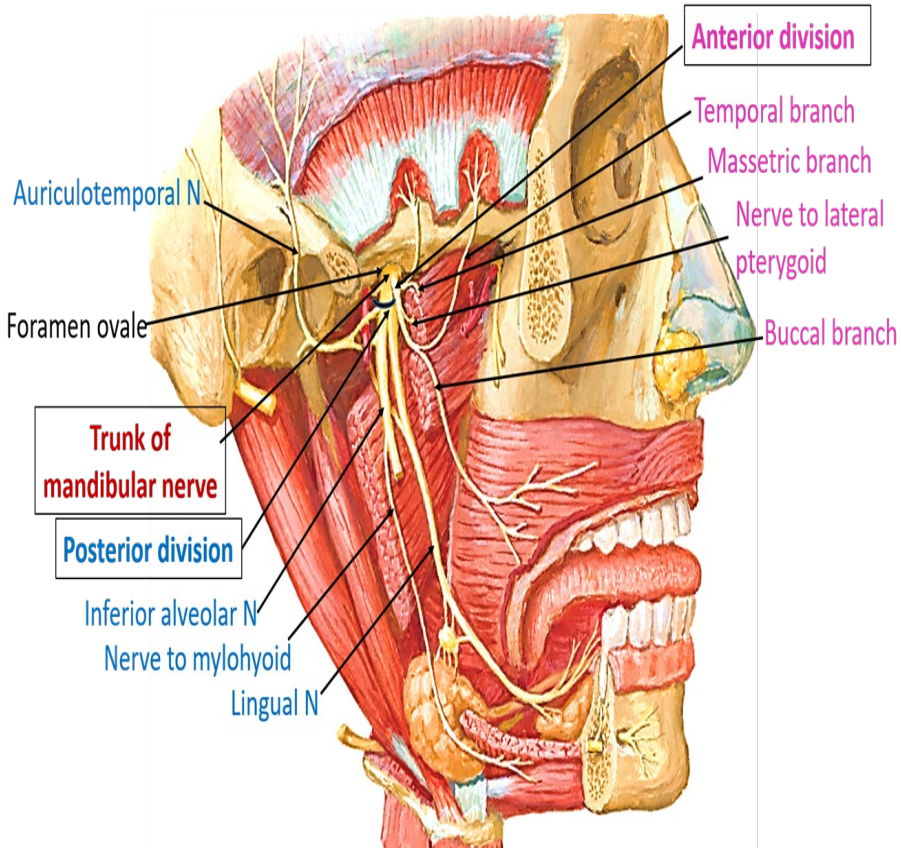
1) Auriculotemporal, 2) Lingual and 3) Inferior alveolar nerves.



Inferior alveolar nerve (course & distribution):

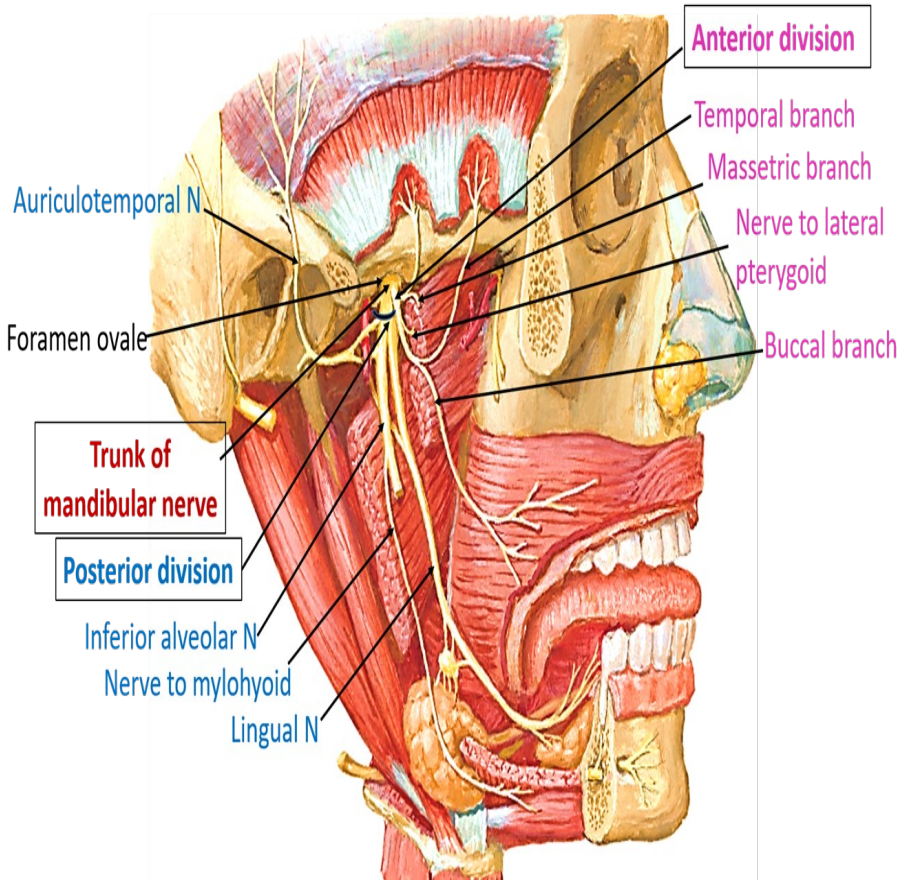
- **Larger terminal branch of mandibular nerve.**
- **Runs vertically downwards to enter mandibular foramen. Runs through mandibular canal and emerges at mental foramen as mental nerve.**
- **Branches:**

a) Mylohoid branch (motor), b) Mental nerve & c) Incisive branch.



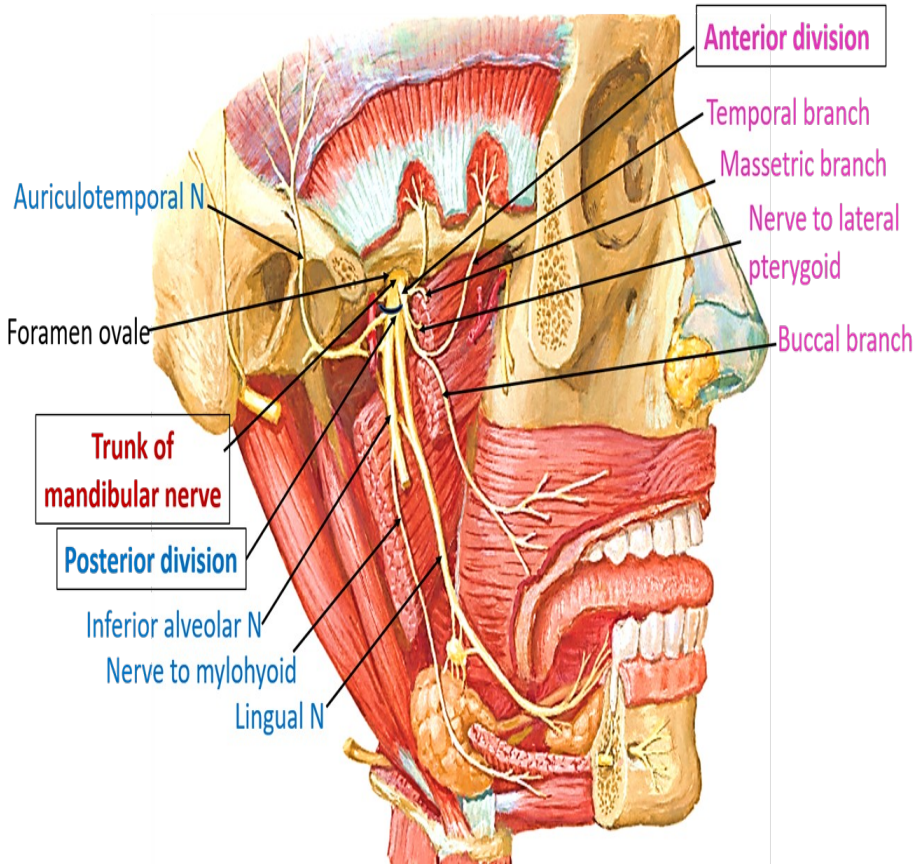
Lingual nerve (course & distribution):

- **Arises about 1 cm below skull.**
- **Joined by corda tympani branch of facial nerve.**
- **Runs downwards & comes in direct contact with mandible medial to third molar tooth.**
- **Finally lies deep to mylohyoid & winds round submandibular duct & divides into terminal branches.**



Auriculotemporal nerve (course & dstribution):

- **Arises by two roots which encircle middle meningeal artery.**
- **Runs upwards behind neck of mandible alongwith superficial temporal vessels.**
- **Distributed as two parts, the auricular part supplies skin of tragus, upper parts of pinna, external acoustic meatus & tympanic membrane and the temporal part supplies skin of temple, parotid gland (secretomotor & sensory) and TMJ.**



Distribution of mandibular nerve:

REFERENCE HANDOUT: TRIGEMINAL NERVE BRANCHES (NOT INCLUDING HITCHHIKING PATHWAYS OF VII, IX) zill@musom.2013

V1 Ophthalmic - GSA (Sensory) only - through Superior Orbital Fissure to Orbit

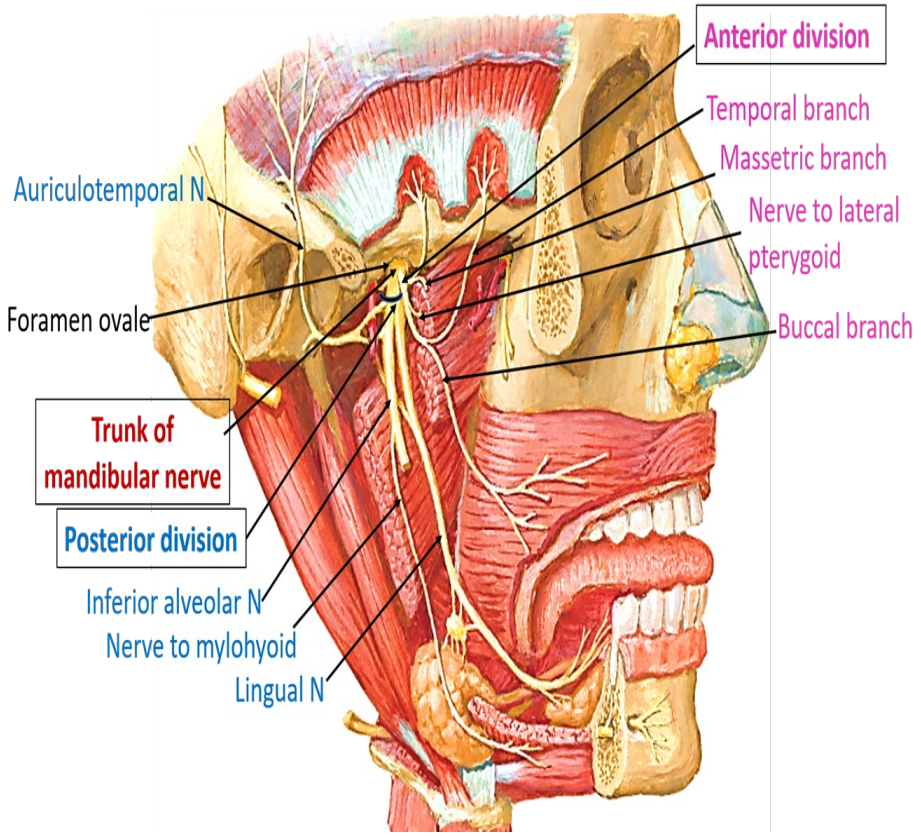
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	b. Supratrochlear Nerve	Scalp forehead, upper eyelid
2. Lacrimal Nerve		Upper eyelid
3. Nasociliary Nerve	a. Long Ciliary Nerve	Cornea of eye
	b. Ant. and Post. Ethmoidal Nerves	Nasal cavity, ethmoid sinus, tip of nose
	c. Infratrochlear Nerve	Upper eyelid, nose

V2 Maxillary - GSA (Sensory) only - through Foramen Rotundum to Pterygopalatine Fossa

Nerve	Branches	Innervates
1. Meningeal branches		Dura of mid. Cranial fossa
2. Ganglionic branches	a. Greater Palatine Nerve	Hard Palate
	b. Lesser Palatine Nerve	Soft Palate
	c. Nasopalatine Nerve	Nasal Cavity, Hard Palate
	d. Nasal branches	Nasal Cavity
3. Post. Sup. Alveolar Nerve		Maxillary teeth
4. Infraorbital nerve		Lower eyelid, nose, upper lip
	a. Ant. Sup. Alveolar Nerve	Maxillary teeth
	b. Mid. Sup. Alveolar Nerve	Maxillary teeth
5. Zygomatic nerve	a. Zygomaticofacial Nerve	Skin of cheek
	b. Zygomaticotemporal Nerve	Skin of temporal region

V3 Mandibular - GSA (Sensory) and SVE (Motor) - Foramen Ovale to Infratemporal Fossa

Nerve	Branches	Innervates
1. Nervous spinosus		Sensory to Dura of mid Cranial fossa
2. Motor branches		Motor to Med. Pterygoid, Tens. Tympani, Tensor Palati
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	c. Inferior Alveolar Nerve i. Nerve to Mylohyoid ii. Mental Nerve	Sensory to Mandibular teeth Motor to Mylohyoid, ant. Digastric Sensory to Chin, Lower lip



Trigeminal neuralgia:

- **It may involve one or more of the three divisions of trigeminal nerve.**
- **Causes severe burning & pain along the distribution of affected nerve.**

Treatment:

1) **Injection of 90% alcohol into affected division of trigeminal ganglion.**
2) **Sectioning the affected nerve or the main sensory root or spinal tract of trigeminal nerve situated in medulla called medllary tractomy.**

Thank you!!